

APPROVED

BSc (Hons) Artificial Intelligence and Data Science
School of Computing, Engineering and Technology

Version	1.0
Course Type	Undergraduate
Award Type	Bachelor of Science with Honours - BSc (Hons)
Awards	
No Course Award Assigned	
Code	0735
SCQF Level	Level 10
QAA Subject Benchmark	Computing (2022)
SCQF Credit Points	480
ECTS Credit Points	240
Valid From	2027 September

HECoS

The maximum number of HECoS Codes allowed is 3.

Code	Title	Alternate Title	Percentage
100359	artificial intelligence	artificial intelligence	50%
100992	machine learning	machine learning	50%
			Total 100%

Professional, Statutory and Regulatory Body (PSRB)

No PSRBs have been recorded

Delivery follows the University Academic Calendar, with two semesters and timetabled sessions during the teaching weeks.

In a typical semester, the student studies four 15-credit modules. The student is normally expected to attend for three hours contact time each week per module. A module will typically feature either a one hour lecture and two hour practical session in the computer labs, or a combined three hour session in the computer labs in which lecturing and student practical activity is interspersed.

The work performed in the practical sessions is selected based on the suitability of techniques for the particular subject covered in a module. It may, for example, consist of the staff

What the Course Involves	supporting the student to engage in problem-solving or may exist as a series of group exercises. The School of Computing, Engineering and Technology has multiple laboratories housing hundreds of computers for the exclusive use of School of Computing, Engineering and Technology students. The student is expected to continue their learning through independent study activities outwith timetabled sessions. Self-directed study is an important aspect of being a student and the student needs to recognise their own learning needs. Typical independent study activities might include directed reading, completion of material from practical sessions, working on a group project, or making progress on a piece of individual coursework. Each semester includes a Connect-and-Reflect Week near the midpoint, which is not used to deliver new academic content, but to provide opportunities for the student to consolidate learning, reflect on progress and development goals, and have a concentrated focus on learning and development priorities. The optional Industrial Placement is an important learning experience. The interaction with the staff of the company, the required tasks and the general environment provides an experience difficult to achieve by other means. In particular, the placement provides the ability to improve soft skills in a way that cannot be meaningfully replicated in the classroom. The Capstone project is an extended piece of work undertaken at the end of the final year. The student agrees a specification for a task, then designs, implements and evaluates an artefact, and produces a finished dissertation under the guidance of a staff supervisor. The subject of the project must be related to the course, and will typically involve the creation of a data system for a particular purpose or the use of artificial intelligence to solve a problem. The entire final semester is devoted to this project and this provides the student with an excellent opportunity to display the abilities they have acquired throughout the course.
UCAS Code	Code G510
Course Collaborations	
No collaborations listed	
Introduction to the Course	Artificial Intelligence and Data Science are two fields which have rapidly expanded and are now seen as critical in many sectors. The former has gained more attention amongst the general public, while the latter is a widely recognised term in industry. In practice, they massively overlap; much Artificial Intelligence is data-driven and requires Data Science skills to enable it to be carried out, while drawing useful insight from data is increasingly being done using Artificial Intelligence. Our BSc (Hons) Artificial Intelligence and Data Science course is designed to introduce the student to modern Artificial Intelligence techniques. Alongside this, the wider practical skill set needed to work effectively with data is provided, allowing the student to be able to develop Artificial Intelligence solutions that can be deployed in the real world. The aim is to produce a graduate with the practical skills needed to fit into a number of different roles within this expanding industry. The course is centred around 3 subject-specific core aspects; the storage and management of data, the extraction of insights from data including using AI, and the presentation of those insights to an end user. These aspects are all introduced in the first year then repeatedly revisited throughout the course, allowing the student to expand their knowledge of the topics while rooting them in skills they have already mastered. In addition, 3 supporting areas are embedded throughout the course to allow the effective use of these skills to solve real problems. First, the study of the legal, ethical, social, professional and security concerns inherent in working with data, which it is crucial for the modern professional to consider. Second, the development of appropriate professional skills, particularly those related to working effectively in teams and in interdisciplinary settings, which are common aspects of the industry the graduate will emerge into. Finally, a strong focus on developing practical computing skills and their application to solving problems, which enables the student to produce working data systems.

	The course introduces the fundamental skills and knowledge in the first year, then expands these to more complex problems and techniques in later years, culminating in a final year that introduces the student to cutting edge, research-driven developments in Artificial Intelligence.
Educational Aims of the Course	Course Aim The overall aim of the course is to provide students with the required skills and knowledge to perform Artificial Intelligence, along with the broader Data Science skills needed to carry out this in real world settings. The course aims to help the student build strong, practical skills across the breadth of the fields of Artificial Intelligence and Data Science, allowing them to pursue a career as a Data Scientist or related profession. As well as the technical focus, the course contains content that aims to prepare the student for the legal, ethical, social, professional, security and business aspects necessary to deploy their skills responsibly in the real world. The course aims to produce a graduate with the soft skills necessary to work effectively in an interdisciplinary team environment. Aims of Each Stage of the Course The first stage aims to introduce the foundation skills and knowledge across the whole breadth of the subject. The second stage aims to advance the skills learned, as well as showing how to bring them together to create a single data system. The third stage aims to complete the broader Data Science education and lay the foundation for the final stage, which aims to cultivate cutting-edge research-led skills focussing on advanced Artificial Intelligence. The course terminates in a semester-long individual Capstone project, which aims to provide the student with an opportunity to apply all they have learned and demonstrate their progress towards being a successful self-directed life-long learner. This is expanded on below. The primary aim of the first year is to introduce the student to the fundamental skills and knowledge that they will need throughout the remainder of the course. The course aims to introduce the student to the fields of Artificial Intelligence and Data Science, and to equip them with introductory skills and knowledge in data storage, extracting information from simple data, and displaying data graphically. The course also aims to get the student thinking about the societal context in which they work in order to develop as a responsible practitioner. Additionally, the course aims to provide the broader foundational computing skills that the student will need for successful progression through the course. The second year aims to expand the student's skills and knowledge in all Artificial Intelligence and Data Science topics covered previously, expanding these to more advanced techniques and more complex data. Another key aim of the year is to show students how to integrate the knowledge from these disparate topics into a single data system. The course-level aim of producing teamworking skills is progressed at this stage through a significant real-world team project. The stage also aims to increase consideration of the real-world and business contexts of data systems for all students through visiting this topic on a number of different modules. The main aim of the course for the student in the third year is to complete their education in the wider field of Data Science, in preparation for a tighter focus on state-of-the-art Artificial Intelligence material the following year. The course also aims to further develop the student's interdisciplinary teamworking skills and the consideration of the business context of their work. The final year of the course aims to provide the student with the final skills they will need to become polished Artificial Intelligence and Data Science graduates. This stage aims to give the student skills in recent, advanced techniques in the field of Artificial Intelligence and to expose them to cutting edge, research-led topics. A final significant Capstone project aims to allow the student to bring together everything they have learned throughout the course and demonstrate their ability to design, implement and evaluate a solution to a sizable Artificial Intelligence and/or Data Science challenge.
Awarding Institution	Robert Gordon University

Course Duration	Full-Time - 4 years, or 5 years if including a year-long industrial placement or a year spend studying abroad.
Distinctive Features of the Course	Personalisation In most semesters, there is the opportunity for the student to select an elective module from the School's wide range of computing modules. This allows the student to explore areas of interest to them and individualise their own learning experience. Optional placement and study abroad years further allow the student the opportunity to choose the university experience that they want. Multidisciplinary Working The first year of all undergraduate computing courses within the School run as a common Foundation Year allowing the student to understand computing as a multidisciplinary subject. In the subsequent two years, students from all undergraduate courses also undertake a module together each semester. These modules allow the student to build their community of practice across the School and develop important skills in working in crossdisciplinary teams. The electives system means that the student will frequently be studying alongside those from other computing courses throughout their studies. The student is encouraged to take part in co-curricula events such as RGUHack, in which they work in a teams with those from other courses to solve industry-set problems. Research-Led Teaching The course is taught by a team of full-time staff who have extensive industrial and research experience in Artificial Intelligence and the wider field of computing. The School has a particular research strengths in Machine Learning and Artificial Intelligence, and the course enables the student to learn from research staff who are active at the leading edge of these technologies. Many of the demonstrators, who assist with the teaching of the course in practical sessions, are PhD students working on Artificial Intelligence and related topics.
Placement/Professional Experience	The student may elect to undertake a short summer placement in a summer break and/or a year-long placement between stages. The placement provides an opportunity to sample working life, be it in commerce, industry or the public sector, and apply knowledge and skills learned in the classroom to the challenges of the workplace. Students who have completed the placement as part of their course may have a definite advantage when it comes to applying for graduate positions. With the combination of training, experience, and a successful degree, students are often offered subsequent employment by the placement provider with which they undertook their placement. While the student is on placement, they will be responsible to a designated member of the placement provider (the Industrial Supervisor) for the discharge of their day-to-day duties. Each student on placement is also assigned an Academic Tutor from the staff of the School, who the student can contact about any academic or non-academic problems during the placement. The Academic Tutor will hold a small number of scheduled video calls with the student and their Industrial Supervisor during the placement. The student may choose to undertake a short summer placement between Stages 1 and 2, or Stages 2 and 3, or Stages 3 and 4. This normally comprises at least 4 weeks of continuous, full-time supervised work experience. The assessment for a short summer placement experience is based on a review of the student's work-related skills, conducted by the Industrial Supervisor. A student on a short summer placement will normally receive at least 1 video call with their Academic Tutor. In addition, or as an alternative to a short summer placement, the student may undertake a year-long placement either between Stage 2 and 3, or between Stage 3 and 4. The year-long industrial placement normally comprises of 30 - 48 weeks of continuous, full-time supervised work experience. The assessment for a year-long placement is based on a review of the student's work-related skills, conducted by both the student and their Industrial Supervisor. A student on a year-long placement will normally receive at least 2 video calls with their Academic Tutor. The University provides support to students interested in undertaking a placement through the Placement Office and Centre for Employability and Community Engagement. The

	Placement Office maintains a pool of placement opportunities and establishes contact with new placement providers as necessary. It will normally assist the student by informing them of suitable placement opportunities. The student is also given help and guidance by the Placement Office in obtaining their own placement from those advertised directly by companies or through their own connections. The Centre for Employability and Community Engagement offers further support for the student, including with preparing their CV and developing interview skills. All placements require approval from the School prior to commencement, in order to ensure that they are suitable for the student and to confirm that the placement provider has put adequate supervisory and health-and-safety arrangements in place. Placements cannot be guaranteed and a student wishing to undertake a placement is expected to reasonably engage with the process, for example by providing a copy of their CV to the Placement Office and by attending any interviews with potential placement providers when required. The credits awarded for the placement module are in addition to the 120 credits achievable from taught modules at each stage. Successful completion is not required for progression to the following stage. The credits achieved for placement may not be used to compensate for any credits from the taught modules.
Opportunities for Further Study	An Honours degree achieved with at least an upper second class (Class 2.1) Classification will normally provide access to postgraduate and doctoral programmes in UK higher education institutions.
External and Internal Indicators of Quality and Standards	SCQF credit values have been used as a guide during the design of modules while the structure and content of the course has also been designed with the British Computer Society's (BCS) requirements for Practical/Engineering Applications in mind, and embedded in the core structure are modules and topics which provide students with computing skills and knowledge in Underlying Theory, Design and Methodology, Tools and Techniques and Project Management and Group work. Standards are primarily established through QAA (Quality Assurance Agency) Benchmarks and the requirements imposed by the British Computer Society in order to gain professional course recognition. The achievement of appropriate standards is demonstrated through the University's course validation procedures and professional institution accreditation visits. The QAA qualification descriptors and Subject Benchmark Statements provide a reference framework that has guided the evolution of the School's degree programme, and the extent of the subject discipline, as defined by QAA, is broadly matched by the provision that now exists within the School's undergraduate programme. A number of internal processes and procedures are used by the School to ensure that standards are maintained and enhanced. Stakeholders' views on standards are routinely obtained from the Staff/Student Liaison Committee, Programme Management Team, Student Voice questionnaires, School Industry Board, industrial visits to placement students, staff involvement in professional body activities and staff participation in accreditation events outside the School and at other HE institutions. In addition to this, best practice in teaching and learning within the University is monitored through the university quality enhancement and quality assurance processes. These include the regular five-yearly cycle of Institution-Led Subject Reviews, as well as participation in the university quality assurance committee structures, such as the University Teaching, Learning and Assessment Sub-Committee. Finally, it should be noted that the lecturers within the School have an ongoing interest in pedagogical research and a commitment to expanding the scholarship of computing education. Academic staff help to organise, and participate in, national and international conferences and have been awarded funding from the Higher Education Academy, the Scottish Informatics and Computer Science Alliance, and the Scottish Enhancement Themes to pursue work in computing education, as well as more general work on student transitions. The School also participates in University conferences on teaching and learning.

University Academic Regulations	This course is governed by the provisions of the University's Academic Regulations, which are available at www.rgu.ac.uk/academicregulations . In particular: Regulation A1: Courses Regulation A2: Admission and Enrolment Regulation A3: Section 1: Student Appeals (Awards and Progression) Procedure Regulation A3: Section 2: Student Conduct Procedure Regulation A3: Section 3: Student Fitness to Practise Procedure Regulation A3: Section 4: Student Fitness to Study Procedure Regulation A4: Assessment and Recommendations of Assessment Boards
University Entry Requirements	Applicants must satisfy the University's general admission requirements to courses as contained in Academic Regulation A2: Admission and Enrolment, including proficiency to a minimum standard in the English language. Specific entry requirements for this course are detailed below.
Course-Specific Entry Requirements	YEAR 1 ENTRY • SQA Higher: BBCC and two further passes to include English and Maths at National 5 grade C or above if not held at Higher. • GCE A Level Year 1 entry: CCC and two further passes to include English and Maths at GCSE grade 5 if not held at A Level. • Irish Leaving Certificate Higher: H3H3H3H3 and two further passes to include English and Maths at grade O3 or above if not held at Higher. • IB Diploma Year 1 entry: 26 to include a pass in Standard Level English and Maths at grade 4 or above if not held at Higher Level. YEAR 2 ENTRY • SQA Advanced Higher Year 2 entry: BBCC to include Computing and another numerate subject and two further passes to include English and Maths at National 5 grade C or above if not held at Higher. • GCE A Level Year 2 entry: BBB to include Math and another numerate subject. English at GCSE grade 5/C is required if not held at A Level. • IB Diploma Year 2 entry: 30 to include Math and another numerate subject at higher level pass. Standard Level English at grade 4 or above if not held at Higher Level. MINIMUM ENTRY REQUIREMENTS The minimum entry requirements for widening access applicants are: • SQA Higher: BCC/CCCC and two further passes to include English and Maths at National 5 grade C or above if not held at Higher.
Alternative Qualifications for Entry	Equivalent qualifications, e.g SQA, BTEC, S/NVQs, overseas qualifications and access programmes will be considered on an individual basis. Applications from mature students with non-standard qualifications or work experience are welcomed.
Interview Procedures	Prospective applicants for the course are not normally interviewed.

Course-Specific Regulations	Honours Classification is based solely on the student's performance in Stage 4 of the course. CM4321 Capstone Project cannot be condoned [refer Academic Regulation A4, paragraph 11]. Electives run at the discretion of the School. Students cannot choose an elective module that they have already taken on the course, for example, some electives are aimed at particular cohorts such as articulating students. Some electives will not be available until the module is first delivered by the School, therefore, please consult the Course Leader for further information on elective availability.
Equality, Diversity and Inclusion	The University has a strong commitment to the active promotion of equality across its functions, including its course provision. To this end the University has developed an inclusivity statement which articulates the University's desire to strive towards creating a working, learning and social atmosphere which is inclusive, harmonious and respectful of diversity. The University has also developed an Equality and Diversity Policy and every two years publishes a mainstreaming report, including equality outcomes, which explains how the university is developing and implementing its inclusivity statement. At a course level, Course Leaders monitor and comment on equalities data through the Annual Appraisal process. Through this process any significant trends in equalities data are highlighted. Further information can be found at: Equality & Diversity RGU. All Schools have a nominated School Contact who can advise on the requirements of the course and the range of course-specific reasonable adjustments that may be made. The University's Inclusion Centre provides support for students, details of which may be found at: The Inclusion Centre: Disability & Dyslexia Support & Advice Services RGU.
University Strategy	The Purpose of the University is: - To be an innovative, inclusive and professionally focused University that has a positive impact on those we serve. The Mission of the University is: - To transform people and communities by providing excellent teaching, research and enterprise opportunities to contribute to economic, social and cultural and environmental sustainability. The Culture of the University is: - To recognise and celebrate the values of curiosity, authenticity, approachability, collaboration, enterprise, ambition, respectfulness, and equality. Driven by the spirit of curiosity to continually improve, our values and principles will guide us. Our partnerships with other organisations and the communities we serve are also a mechanism to develop and grow our diversity. We have a strong commitment to widening and extending access to our courses, regardless of an individual's background. At the core of the University's significant achievements is the commitment and dedication of our people. The sense of partnership between our staff and students is strong and supporting our students to succeed in their studies and beyond is deeply ingrained in our culture. These features are reflected in the University's student satisfaction rates, which in recent years have consistently placed RGU in the top 10 of universities in the UK. RGU has a long history of inspiring highly employable graduates. This is achieved through close, collaborative contact with employers in the design and delivery of our curriculum; the range and breadth of accreditations with professional and statutory bodies; the significant role work placements and work-based learning play within our courses; and the embedding of

	enterprise across all our activity. In a changing society, we will ensure the continued evolution of our curriculum so that our graduates can continue to succeed in their ambitions and make a positive impact within society well into the future. To achieve our ambitions, our curriculum must be global in outlook, and we must maximise the international reach of our course portfolio. The University's research activities must have international impact and our enterprise activities must reflect the global reach of the economic sectors we work with and the ambitions of the region. RGU recognises that the creation of a sustainable future for all is a priority that unites us globally, and which demands expertise, creativity, and innovation from all subject areas and professional disciplines. Consequently, sustainability is integrated into all courses, equipping students with skills and competencies that are essential for their future careers and lives.
	The BSc (Hons) Artificial Intelligence and Data Science course has been specifically designed to create graduates with the necessary practical skills and abilities to commence a career as a Data Scientist or a related profession such as a Data Engineer or Machine Learning Engineer. The course emphasises the development of strong, practical computing skills that can be used to solve real problems. Authentic assessment is used as widely as possible, requiring students to create computing artefacts or analyse data. Industry standard software and tools are widely used to build up the competencies needed to transition into employment on graduation. Within the course there is an adherence to three principles: • To encourage innovation through investigation and understanding of Artificial Intelligence and Data Science and related computing technologies. • To practice the process of information system design and development, across multiple platforms and programming languages. • To demonstrate the practical application of skills and knowledge in Artificial Intelligence and Data Science to encourage and support creative development and problem-solving. The course is structured around the development of 3 core subject-specific skills, and 3 supporting skills. Subject Specific Skills • Data Storage and Management The collection, storage and management of data is a necessary first step in any project in the area. The course provides the student with the opportunity to learn about a large range of different technologies in this area, suitable for different needs and different types of project. This ensures that the student has a range of options at their disposal when designing and creating systems. Wherever a graduate ends up in the field, they will have encountered the key data storage tools and techniques needed in that domain. • Extracting Insights from Data Deriving useful information from raw data is one of the key aspects of a successful information system. From beginning to end of the course, the student will learn techniques to do this with increasing complexity. From basic mathematics through to modern deep learning approaches, the student will acquire the practical skills needed to analyse data and interpret their findings. The course has a strong emphasis on acquiring the skills needed to perform these tasks for real with industry standard languages and tools, rather than restricting itself to focusing on theoretical knowledge. • Data Visualisation Presenting insights from a dataset to a range of different users with different backgrounds and understanding is essential if those insights are to be useful. The course includes a human-computer interaction (HCI) theme which gives the student the knowledge to consider their end users and tailor their outputs for them. Again, this is introduced in a very practical way, with the student learning to use modern tools and technologies to create user interfaces and present data graphically. Crossdisciplinary work and work with real-world clients are features of the course and give the student further opportunity to understand interaction with those who have different views and understanding of a problem. Supporting Skills

Course-Specific Learning Experience

- **Legal, Ethical, Social, Professional, Security and Business Considerations** Computer systems do not operate in a vacuum and the modern AI and Data Science professional must consider the context in which they are working. The students is taught about the principles involved in these areas, both as a stand alone topic and through threading them into other modules where the issues arise. The business case for Data Science projects is considered including in the context of entrepreneurialism in stage 3. The students is expected to learn to question what is legal and ethical when designing and implementing systems, and these skills are particularly exercised in the real world projects and final Capstone project.
- **Professional Skills** Professional skills, such as project management, planning, communication, presentation, and writing skills are extremely desirable by industry. A number of modules over the course provide the student with skills in these areas and opportunities to demonstrate them so they are fully prepared to enter the workforce on graduation. In particular, there is a focus on developing the skills needed to work in a team and communicating with others from different disciplines; A graduate in this area will often be the expert in their topic and must be able to work effectively with others.
- **Technical Computing Skills** The course has a strong focus on building working artefacts and analysis of real data sets. This requires the development of foundational computing skills, such as programming and web development. The student will be taught these skills in earlier years, which will support them to create subject-specific applications.

The course is designed around a 'spiral curriculum', in which all the core areas above are introduced in the first year then revisited and further built on throughout the stages. This enables students to encounter the key topics in the field early and to relate new learning to what they already know. For example, to extract useful information from data, students start with the simplest of tasks - calculating an average. This then builds in subsequent modules to using Data Analysis, then Data Mining, then Machine Learning, and finally modern Deep Learning. Similar expansions of knowledge are present for the other aspects mentioned above.

The course culminates with the Capstone project. This allows the student to show a full, complex piece of work they have designed, documented and implemented using skills from throughout the course, and which they can be proud of producing.

Personalisation

The course allows the student to tailor the course to their particular interests and career aspirations. In most semesters, there is the opportunity for the student to select an elective module from the School of Computing, Engineering and Technology's range of computing modules. This allows the student to explore areas of interest to them beyond the core topics for the course and hence individualise their own learning experience. The opportunity for placement and study abroad years further allow the student the chance to choose the university experience that they want.

Multidisciplinary Working

The first year of all undergraduate computing courses within the School of Computing, Engineering and Technology run as a common Foundation Year allowing the student to understand computing as a multidisciplinary subject. In the subsequent two years, students from all undergraduate courses also undertake a module together each semester. These modules allow the student to build their community of practice across the School and develop important skills in working in interdisciplinary teams. The electives system means that the student will frequently be studying alongside those from other computing courses throughout their studies. The student is encouraged to take part in co-curricula events such as RGUHack, in which they work in teams with those from other courses to solve industry-set problems.

External Competitions and Challenges

As part of the course, the student is encouraged to take part in various co-curricular competitions and challenges throughout the year. In recent years, these challenges have included:

- The School's "Hackathon" (RGUHack), where students work in teams for around 24 hours on challenges set by industry partners.
- RGU GameJam, which is an opportunity for students to develop a computer or board game within 30 hours based on a given theme.
- The Google Hash Code challenge, where students work on large optimisation challenges set by Google over a period of four hours.
- The School has also sent teams of students to compete in the UK Cyber Security Challenges.

The School strongly believe, and student feedback has shown, that competing in these events raises the student's confidence in their own ability and allows them to use the knowledge and skills they have learned in a pressured, but consequence-free, environment.

Industry Visits

The School prides itself on its close links to industry; As part of this we arrange a number of opportunities for industry partners to both visit the School and host visits on their premises for the student.

Technical Facilities

Many of the practical aspects of the course take place in the computer laboratory. The School of Computing, Engineering and Technology has multiple laboratories housing hundreds of computers, equipped with the latest versions of open source and commercial software for the exclusive use of the School's students. Throughout the School, there is 100% wireless coverage that allows access to the University network for the student. The computers in the School's labs are normally updated on a regular replacement cycle to ensure that the School's computing facilities are capable of delivering high performance and can run the latest software. Access to the labs is by electronic key fob at any time during University opening hours. Software provision is comprehensive, including programming environments for most modern programming languages, database systems including Oracle, SQL Server and MS Office packages. Much of the software used on the course is open source or covered under an education license, allow the student to continue their work at home using the same set up as available in the labs.

Sustainability

The course includes sustainability in 3 ways.

In the first semester of the first year, the student studies the module CM1122 AI, Data and Society. This encourages them to think about the potential of what they learn on their course to be used positively and also the negative impacts that may follow from misuse of their skills. Very early inclusion of this topic means the student should be aware of the importance of the topic and question the appropriate usage of skills as they meet them in later stages.

Throughout the course, sustainability is embedded through the inclusion of relevant case studies in modules where appropriate. For example, students may be asked to reflect on whether it is acceptable to use a dataset for a specific machine learning task, or asked to produce a computer program addressing a particular sustainability challenge

Capstone projects frequently include those influenced by sustainability concerns. Recent relevant projects in the School have included using Machine Learning to predict pollution levels and investigating applying Data Science methods to Environmental, Social and Governance data.

The Learning Process: Teaching Strategy and Methods

The course curriculum is designed as a coherent whole with an emphasis on the integration of theory, practical skills and applications. It aims to enable the student to acquire and develop the subject knowledge and understanding necessary for successful professional employment and supplies methodologies for the adoption of a disciplined approach to tasks such as problem definition, specification, design, implementation, and maintenance. Such knowledge includes an understanding of the range of possible options for these tasks, suitable technologies, the related practical and transferable skills including relevant approaches to collaborative work, and access to the appropriate software tools and resources. It also provides the necessary underpinning to guide practice, to ensure the sustainability of their knowledge and to provide an appropriate framework that will accommodate rapid technological change.

These skills are developed at the Degree and Honours levels corresponding to the Scottish Credit and Qualifications Framework levels 7 to 10.

A general feature of the course is that it stresses a unified, integrated view of the subject area.

This is reflected in the fact that it is designed to:

Teaching and Learning Strategy

- Present a coherent underpinning of the theory appropriate to the aims of the course which is further developed and used throughout the period of study. This enables the graduating student to adapt to future developments in the field.
- Reflect the rapid rate of change in the field and ensure that coverage is given to a selection of emerging topics so that the student is aware of likely future developments in the subject together with their potential impact.
- Integrate the delivery of teaching across thematic boundaries to highlight fundamental aspects of interconnectivity and interdisciplinarity within different areas of the syllabus.
- Integrate modes of assessment to attempt to balance the types of assessment (individual and group) and purposes of assessment (e.g. formative and summative).
- Facilitate the student to exhibit competence and achievement across a number of different themes (integrated assessment).
- Encourage the student to combine elements of their learning from different parts of the course and to show their accumulated and integrated knowledge and understanding of a topic or subject area (synoptic assessment).
- Allow the student to study elective topics from across other fields of computing to explore areas of interest and better understand the relationship between Artificial Intelligence and Data Science and the wider context in which it occurs.

The professionally orientated nature of the course requires the student to obtain general and specific skills, this being addressed by a combination of teaching and learning methods, including lectures, tutorials, computing laboratories, workshops, seminars, practical work, optional placement, integrative project work, student-centred learning and private study.

Teaching and Learning Approaches

The main goal of the teaching and learning strategy is to enable the student to become a critical, reflective, independent learner who is an active participant in the learning process and who develops the motivation and autonomy needed to take control and regulate their own educational experiences. This means that, throughout their course of study, the student will begin to take more responsibility for their own acquisition of necessary skills and knowledge so that, by the end of their course, they will be in a position to evaluate and assess their own effectiveness.

This process is driven by various objectives such as the need for constructive alignment of syllabus and learning approaches and makes use of a wide range of appropriate delivery styles. These include conventional instructional teaching methodologies, social learning activities such as group and team-working, as well as constructivist, investigative approaches such as problem-based learning and enquiry-based learning.

As well as the day-to-day use of the latest computer hardware and software afforded by the degree, extensive use of technology-assisted learning is made within the course of study. Examples of these include assessment by means of computer-assisted assessment packages embedded in the virtual learning environment to engage the student and increase the speed of feedback and the use of an integrated approach whereby students have short periods of instruction followed by immediate lab exercises to contextualise the topic.

Modes of Delivery

Lectures

This is usually a formal discourse for the purposes of dissemination of information, the demonstration of techniques where appropriate and the discussion of supporting ideas and consequences. The lecture is supported by a network of dissemination equipment including computer projection facilities, whiteboard, and video where appropriate. Although this type of presentation is suitable for a one-sided discourse ample opportunity exists for questions, interaction and discussion and this is facilitated by technological means.

Tutorials

Within the School, this format is used for a wide range of activities, each suited to the particular subject. It may consist of the staff supporting students engaged in problem-solving or may exist as a series of group exercises where each group is encouraged to allocate responsibilities, allocate tasks, etc. Generally this type of teaching is used to support the lecture, clarify the material and experiment with the techniques and skills required.

Laboratories

The nature of the computing elements of all courses requires students to gain practical skills in the use of a computer PC or workstation. This activity takes place in one of the School's computer laboratories and consists of the student, supported by staff members, practising skills in the use of sophisticated software applications and including software development and systems analysis and design tools. There has also been increasing emphasis on using an approach to subjects in which a series of short mini-lectures are delivered in labs to allow the student to gain theoretical knowledge of a topic. These are interspersed with selected practical exercises, which allow this knowledge to be immediately assimilated and so reinforce the learning process.

Industrial Experience

The course includes a placement, which is a useful supplement to the other teaching and learning activities of the course. A student participating in the Industrial Placement has to competitively apply for placements by submitting letters of application and CVs and by attending interviews. A student on placement undertakes a variety of roles and tasks for the company they are working for, supported on the placement by an Academic Tutor from the staff of the School and an Industrial Supervisor from the company. The interaction with the staff of the company, the required tasks and the general environment provides an experience difficult to achieve by other means. The Academic Tutor supports and monitors the assessment process. In addition to the Industrial Placement, there is a series of presentations and seminars given by academic staff, staff from the careers advisory service and professionals working in key positions within the Computing and Information Technology sectors of business and industry. These presentations and seminars form a key part of the CM2112 Real World Projects and Professional Skills module.

Group Projects

These are extended pieces of work, completed under supervision of a staff tutor. The student is assigned to a small project team and given a specification for a task, which may be the analysis, design and partial implementation of an information system, the development of software or similar. The group manages themselves and work as an organised team to produce a set of deliverables, which includes the design and development of software and accompanying documentation to professional standard.

Individual Project

This is an extended piece of work, completed under supervision of a staff tutor. The student agrees a specification for a task, which may be the analysis and design of a system, the development of software or similar and produces a finished thesis and product. This activity replicates the development process that would occur in an industrial development or research setting.

The Student's Role in Learning

The student is expected to be active participants in the learning process. This means that there is a requirement for the student to extend their learning from formally delivered materials through wider reading, research, practice and self-directed enquiry. As the student develops over the duration of the course, they are expected to become increasingly self-directed and self-motivated, taking greater control of the study process, and defining their personal objectives for learning. The student is encouraged to reflect regularly upon their learning and to establish a view of their own progress and development, which may form the basis of tutorial guidance with staff. The student is expected to utilise the full range of facilities and resources available to them, for example the services and resources provided by the library.

The Assessment of Student Work

The University has developed a Grading Scheme for assessment purposes. A variety of assessment methods are used in this course, as deemed appropriate to the subject material being assessed and the context in which the material is delivered. It is important to note that all grades indicated to the student during the course of the academic session are provisional until the deliberations of the Assessment Board for the course. Provisional grades are presented to this Board, which consists of academic staff involved in course delivery as well as External Examiners, and a final decision is confirmed on the performance of every student for each module undertaken by them. This is subsequently confirmed to each student in a published transcript, and may be accessed electronically on the Student Portal.

Assessment

Work-Based Learning

The optional placement is assessed by a combination of Tutor and Supervisor assessment of the student's work, based on direct observation of their performance and a report submitted by the student.

Computer Aided Assessment

Computer-assisted assessment packages integrated into the virtual learning environment such as Moodle's question system are used in many modules, both for formative and summative assessment, to enhance student engagement and increase the speed of feedback.

Peer Assessment

Peer Assessment is utilised in many modules throughout the course where group work is undertaken. This serves to develop working and interpersonal skills, as well as honing appraisal and critique skills. Peer assessment may be used in both summative and formative contexts.

Unseen Exams

In general unseen exams are used to assess students' knowledge and understanding.

Assessed Laboratories

Assessed Laboratories are used to assess the student's knowledge, understanding and practical skills, and generally involve the student undertaking an unseen practical task such as developing a part of an information system, under closed book, supervised conditions with a fixed time constraint.

Coursework

Coursework may take many forms, i.e. reports, essays, production of software, design documentation and modelling activities. The School prepares a schedule of coursework assessments for each teaching semester in order to balance the student's workload.

Projects/Dissertations

Extended projects are used in later stages as a learning vehicle and assessment process that allow the student to tackle more complex problems that often require the integration of knowledge and understanding, and intellectual skills from a variety of subject areas. These projects often require the practical application of a number of different technologies and provide the student with the opportunity to learn new investigative and evaluative skills. Projects, in particular group projects, provide an opportunity to learn new investigative and evaluative skills, and are also often used as the vehicle for gaining valuable employability skills. Project work is usually assessed by a combination of assessment of the completed project deliverable as an artefact, and an accompanying report or dissertation documenting the project work undertaken by the student.

Assessment Types

Modules may use a combination of assessment types that integrate modes of assessment in order to balance the types of activity (e.g. individual and group activities) and purposes of assessment (e.g. formative and summative).

Where there are a number of modules that cover allied subject material integrated assessment may be used to facilitate the student exhibiting competence and achievement across a number of different themes.

In all years, synoptic assessments embedded into projects and other practical assignments are used to encourage the student to combine elements of their learning from different parts of the course and to show their accumulated knowledge and understanding of a set of topics or a subject area.

The student's Honours Classification (First Class Honours, Upper Second Class Honours etc) is based solely on performance in Stage 4 of the course.

	Equality and Diversity Any datasets used within the course will aim to adhere to the Software Engineering Code of Ethics and Professional Practice recommended by the ACM/IEEE-CS Joint Task Force which states that "only accurate data derived by ethical and lawful means" will be used, and that its integrity will be maintained. Furthermore, the student will be taught about recognising and researching known bias within datasets and to consider legal, ethical, social and professional issues in their work.
University Support for Teaching, Learning and Assessment	The University provides a number of means of supporting teaching and learning: • student induction organised on a course or School basis; • an ongoing scheme of personal/pastoral support for students, including online support, provided by the <i>Inclusion Centre</i>; • an extensive programme of student study skills delivered through <i>Student Life</i> and the <i>Library</i>; • an extensive library of learning resources; • close collaboration with industry and professional, statutory and regulatory bodies; • the University's active participation in the Quality Assurance Agency's Quality Enhancement Themes, enhancementthemes.ac.uk ; • support for staff from the <i>Department of Academic Quality and Learning Enhancement (AQLE)</i>, including credit rated teaching, learning and assessment provision; • extensive opportunities for student placements with companies or organisations, academic overseas exchange programmes, and support for entrepreneurial activity and community engagement opportunities; • access to a range of professionally qualified, quality assured careers professionals to support individual students and graduates career planning and opportunity search, supported by a range of bespoke and contemporary online advice and information including the extensive RGU eHub; • <i>Moodle</i>, http://campusmoodle.rgu.ac.uk/ , the University's dedicated virtual learning environment; • a commitment to knowledge exchange and technology transfer through focused research activity, which contributes to the critical underpinning for all taught courses; • the expanding provision of state-of-the-art, purpose-built facilities and buildings.
Course-Specific Support for Teaching, Learning and Assessment	In addition to the University-wide support for Teaching and Learning, the School of Computing, Engineering and Technology provides a number of means of supporting the student in teaching and learning. Members of staff publish course-based teaching and learning materials on the University's Virtual Learning Environment, Moodle. The range of published materials spans from supplementary teaching materials such as additional lecture notes and tutorial guides to software development to general guidance for the student, such as codes of conduct in laboratories. Information on studying abroad can be accessed at https://www.rgu.ac.uk/study/study-abroad Feedback is provided continuously to the student through informal contact with subject lecturers and tutors in the tutorial and laboratory settings. Feedback is provided on assessed practical work normally within 20 working days of submission of the work. Facilities are made available for the student to view their marked examination scripts. Grades are made available electronically for viewing via the student portal following ratification by the Assessment Board.
	The University employs several mechanisms for evaluating and improving the quality and standards of teaching, learning and assessment, including: • <i>Annual Course Appraisal Reports</i> are prepared for each course and reviewed and approved by Course/Programme Management Teams and School Academic Boards which consider, amongst other things, feedback generated from student questionnaires;

University Monitoring of Quality and Standards	• <i>Institution-Led Subject Review</i> , involving external panel members, on a six-yearly basis to formally review its major subject provision, followed by a three year interim review to monitor progress against actions/issues raised through the review process; • <i>External Examiner Annual Reports</i> ; • Ongoing liaison with industrial/professional liaison groups. Formal Committees with responsibility for monitoring and evaluating quality and standards: • Staff/Student Liaison Committees, or equivalent; • Course/Programme Management Teams; • School Academic Boards; • Assessment Boards; • Quality Assurance and Enhancement Committee and associated Sub-Committees.
Course-Specific Monitoring of Quality and Standards	Annual Course Appraisals are reviewed and approved by the Programme Management Team and School Academic Board which meets at least once per semester. The Programme Management Team considers teaching and learning issues arising from feedback generated by External Examiner Annual Reports, Staff/Student Liaison Committees, and Student Voice Questionnaires. The Programme Management Team and School Academic Board also review all proposed changes to the structure and content of the course. The course is also informed by feedback from the School's Industrial Board.
University Student Feedback Mechanisms	The University and RGU:Union work in partnership to create an environment which stimulates: • the participation of students, by empowering them to proactively provide views and opinions; • a supportive learning community where students and staff engage in meaningful dialogue; and • the engagement of students in the design of solutions and enhancements. This approach is underpinned by the <i>Student Partnership Agreement</i> which promotes the ethos of partnership at all levels of the University. Further information can be found at the Student Representation and Partnership Moodle Area: Student Representation & Partnership RGU . There are a variety of opportunities for students to provide feedback to the University, and to become actively involved in shaping their learning experience. These opportunities are integral to the University's approach to the quality assurance and enhancement of teaching and learning, and the holistic student experience. Mechanisms through which students are engaged and supported include: • participation in student evaluation questionnaires and where relevant the <i>National Student Survey (NSS)</i> ; • staff/student liaison arrangements at course/programme level;

	• support for Student Representatives; • student representation on Institution-Led Subject Review Panels and Validations; • student representation on key University committees.
Course-Specific Student Feedback Mechanisms	There is a joint Staff/Student Liaison Committee for all undergraduate degree courses. It normally consists of the Undergraduate Programme Leader and undergraduate Course Leaders, with student representatives from each stage of each course. The Committee meets at least once per semester and feeds appropriate information to other departments and/or committees as required. Student representatives also take part in the School Academic Board where they can contribute to discussions about School policy. In addition, there are regular scheduled meetings between Course Leaders and student representatives to discuss issues arising within specific classes and on individual modules. Student Voice Questionnaires are distributed to the student for all modules midway through the semester, which allows them to raise issues and give feedback to staff.

Awards

Stage 1

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Certificate of Higher Education - CertHE	Artificial Intelligence and Data Science	Aberdeen, United Kingdom	Exit Award		120

Stage 2

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Diploma of Higher Education - DipHE	Artificial Intelligence and Data Science	Aberdeen, United Kingdom	Exit Award		240

Stage 3

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Bachelor of Science - BSc	Artificial Intelligence and Data Science	Aberdeen, United Kingdom	Exit Award		360

Stage 4

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Bachelor of Science with Honours - BSc (Hons)	Artificial Intelligence and Data Science	Aberdeen, United Kingdom	Target Award		480

Level Learning Outcomes

On successful completion of this Course the learner will be able to :

#	Description	SCQF Characteristic	SCQF Level
LLO1	Describe the main content and context of Artificial Intelligence and Data Science, and their relationship to each other and other subject areas such as computer science, mathematics, science and engineering.	SC1	SCQF 7
LLO2	Outline familiarity with the basic facts and principles of computing and supporting topics via demonstrations of skills gained, the ability to answer questions and to describe and employ basic tools and techniques.	SC1	SCQF 7
LLO3	Describe the basic principles of programming and the development of software.	SC1	SCQF 7
LLO4	Recall basic information about the principles and techniques involved in data storage, analysis and dissemination.	SC1	SCQF 7
LLO5	Recognise legal, ethical, social, professional, security, environmental and business issues occurring in a computing context.	SC1	SCQF 7
LLO6	Carry out study of the subject in a systematic, directed way with the aid of appropriate tutor guidance within the classroom and laboratory.	SC2	SCQF 7
LLO7	Apply the knowledge, skills and understanding of programming, data storage, analysis and dissemination in practical contexts.	SC2	SCQF 7
LLO8	Employ some of the basic and routine professional skills, techniques, and practices associated with computing in both routine and non-routine contexts.	SC2	SCQF 7
LLO9	Apply the basic principles and procedures found in computing to standard, simple situations.	SC3	SCQF 7
LLO10	Present analysis of situations and problems, based on the set of classifications presented by tutors.	SC3	SCQF 7
LLO11	Relate a limited number of facts/ideas/elements in an imitative manner.	SC3	SCQF 7
LLO12	Choose their own value judgements based on criteria provided.	SC3	SCQF 7
LLO13	Apply the basic tools/methods of computing to simple, standard problems.	SC3	SCQF 7
LLO14	Convey aspects of computing in a clear and concise way, in writing and orally, in relatively informal and limited-length pieces of work.	SC4	SCQF 7
LLO15	Present appropriately the basic design and documentation of programs.	SC4	SCQF 7
LLO16	Obtain basic numeracy, mathematics and the ability to manipulate data at a level appropriate to this stage of the degree.	SC4	SCQF 7
LLO17	Achieve an understanding of basic concepts in logic, statistics and computation.	SC4	SCQF 7
LLO18	Present practical skills in computer science using basic ICT applications as well as those skills necessary for efficient use of computer networks.	SC4	SCQF 7
LLO19	Learn in an increasingly effective and purposeful way, showing the beginnings of development as an autonomous learner, and exercise some initiative and independence in carrying out defined activities at a professional level of practice.	SC5	SCQF 7
LLO20	Interact effectively with tutors and fellow students via software and web development tasks, problem-solving tutorials and the production of appropriate documentation, and accept supervision in less familiar areas of work.	SC5	SCQF 7
LLO21	Exercise some managerial or supervisory responsibility for the work of others within a defined and supervised structure while managing limited resources within defined areas of work.	SC5	SCQF 7
LLO22	Take the lead in implementing agreed plans in familiar or defined contexts, while taking account of own and others' roles and responsibilities when carrying out and evaluating tasks.	SC5	SCQF 7
LLO23	Work, under guidance, with others to acquire an understanding of current professional practice.	SC5	SCQF 7
LLO24	Report a growing appreciation of the scope of the field of study so enabling students to map knowledge into a coherent picture of the discipline.	SC1	SCQF 8

LLO25	Report an appreciation of the significance of central and peripheral areas of subject, suggesting opportunities to explore and extend the boundaries of the discipline.	SC1	SCQF 8
LLO26	Compare the suitability of alternative techniques involved in data storage, analysis and dissemination for a range of purposes.	SC1	SCQF 8
LLO27	Report on the legal, ethical, security and business concerns that apply to a Data Science project.	SC1	SCQF 8
LLO28	Show knowledge, skills and understanding in a range of professional skills, techniques, and practices associated with the topics studied, some of which are at an advanced or complex level.	SC2	SCQF 8
LLO29	Use routine lines of enquiry, development or investigation into professional level problems and issues.	SC2	SCQF 8
LLO30	Adapt routine practices within accepted standards.	SC2	SCQF 8
LLO31	Use a variety of tools and techniques for the creation of software for data storage, analysis and dissemination.	SC2	SCQF 8
LLO32	Use standard principles and procedures in somewhat more demanding situations.	SC3	SCQF 8
LLO33	Combine a range of facts, ideas and elements in an argued case.	SC3	SCQF 8
LLO34	Distinguish between a range of valid alternative solutions in a reasoned, systematic and increasingly independent way, particularly within the context of group work.	SC3	SCQF 8
LLO35	Use given tools and methods accurately and carefully in more demanding problems appropriate to this stage.	SC3	SCQF 8
LLO36	Use communication skills in more formal academic and professional styles, and in more extended pieces of work.	SC4	SCQF 8
LLO37	Undertake a selection of more advanced numerical, modelling and design tasks, relevant to data mining, machine learning and data storage.	SC4	SCQF 8
LLO38	Use a broad selection of more advanced computing and ICT skills in a wide range of contexts relevant to topics at this level.	SC4	SCQF 8
LLO39	Practice effective work within a learning- or subject-specific group, demonstrating basic negotiating, role-ownership, leadership and group-support skills.	SC5	SCQF 8
LLO40	Plan work activities in familiar or defined contexts.	SC5	SCQF 8
LLO41	Practice teamwork in ways that show awareness of their own and others' roles, responsibilities and contributions.	SC5	SCQF 8
LLO42	Compare attitudes and approaches to handling legal, ethical and professional issues with others to form an agreed joint plan of action on a task.	SC5	SCQF 8
LLO43	Explain a widening range of concepts, theories and techniques used in data storage, analysis and dissemination.	SC1	SCQF 9
LLO44	Discuss the broad areas of the knowledge/subject bases of computing including specialist areas.	SC1	SCQF 9
LLO45	Review the limits of the broad area of study, together with an ability to relate the different areas to one another in a systematic and logical manner.	SC1	SCQF 9
LLO46	Explain the key principles involved in artificial intelligence and its application to solving problems.	SC1	SCQF 9
LLO47	Demonstrate knowledge, skills and understanding in using a range of the principal professional skills, techniques, practices and/or materials associated with computing.	SC2	SCQF 9
LLO48	Demonstrate an applied knowledge of some specialised or advanced skills, techniques, practices and/or materials in the implementation of computer systems solutions.	SC2	SCQF 9
LLO49	Demonstrate routine methods of enquiry and research, including in a range of professional contexts that include a degree of unpredictability.	SC2	SCQF 9
LLO50	Demonstrate the ability to apply artificial intelligence techniques to a range of problems.	SC2	SCQF 9

LLO51	Demonstrate the ability to select and use principles and procedures appropriate to the situation or problem in hand.	SC3	
LLO52	Draw on familiar ideas or principles in new contexts or situations to analyse effectively.	SC3	SCQF 9
LLO53	Assemble facts, ideas and elements into a well argued case and to produces new design ideas.	SC3	SCQF 9
LLO54	Discriminates between a range of design ideas in a reasoned, systematic and increasingly independent way.	SC3	SCQF 9
LLO55	Demonstrate more self-direction in identifying key elements or problems and choosing appropriate methods to construct their resolution, with only general guidance.	SC3	SCQF 9
LLO56	Judge appropriate methods for communicating reports and presentations on relevant aspects of computing to different audiences.	SC4	SCQF 9
LLO57	Defend the selection of specialist numerical, modelling and design skills, relevant to the development and implementation of information systems.	SC4	SCQF 9
LLO58	Assess the appropriate use of a broad selection of specialist computing and ICT skills in a wide range of contexts.	SC4	SCQF 9
LLO59	Demonstrate a broad-ranging and flexible approach to study, identifying learning needs and pursuing activities designed to meet these needs in increasingly autonomous ways.	SC5	SCQF 9
LLO60	Seek out appropriate roles in interdisciplinary team projects.	SC5	SCQF 9
LLO61	Judge appropriate techniques for negotiation and collaboration with those from other computing disciplines and from outside the world of computing.	SC5	SCQF 9
LLO62	Judge the most appropriate way to handle legal, ethical and professional issues in accordance with current codes of best practice, seeking guidance where appropriate.	SC5	SCQF 9
LLO63	Develop an understanding of the context of, and relationship between, the principal areas, features, boundaries, terminology and conventions in data storage, analysis and dissemination.	SC1	SCQF 10
LLO64	Conceptualise the specification, design, implementation and evaluation of a substantial computing project, using appropriate high-level expertise.	SC1	SCQF 10
LLO65	Offer a clear understanding of their subject's boundaries, of the limits of its application, and of the range of methods of study and types of judgment employed.	SC1	SCQF 10
LLO66	Offer a detailed knowledge and understanding of artificial intelligence, informed by research at the forefront of the subject.	SC1	SCQF 10
LLO67	Justify the selection of artificial intelligence techniques in the solution of a range of diverse applications in the field.	SC1	SCQF 10
LLO68	Execute knowledge, skills and understanding in using a wide range of the principal professional skills, techniques, and practices associated with computing.	SC2	SCQF 10
LLO69	Execute practical skills in the use of techniques and materials that are specialised, advanced and at the forefront of the subject of Artificial Intelligence and Data Science	SC2	SCQF 10
LLO70	Execute a defined project of research, development and investigation in a substantial project though the application of a practical knowledge of computing, Artificial Intelligence and Data Science.	SC2	SCQF 10
LLO71	Operate in a range of professional level contexts that include a greater degree of unpredictability and specialism.	SC2	SCQF 10
LLO72	Operate with complex ethical and professional issues in accordance with current professional practice.	SC2	SCQF 10
LLO73	Illustrate the use of a range of both data-driven and model-based artificial intelligence techniques to solve various problems.	SC2	SCQF 10

LLO74	Execute the use of artificial intelligence techniques subject to a range of legal, ethical and professional issues.	SC2	SCQF 10
LLO75	Argue for the correct selection and application of principles and procedures to the solution to a range of professional situations and problems while working autonomously.	SC3	SCQF 10
LLO76	Critique principles and ideas systematically and effectively in new contexts and situations.	SC3	SCQF 10
LLO77	Compose facts, ideas and elements in support of a well-structured argument or to design novel solutions and evolve new concepts, while working autonomously.	SC3	SCQF 10
LLO78	Reconcile theory with professional and vocational practice to evaluate theory, process, solutions and outcomes critically and effectively.	SC3	SCQF 10
LLO79	Conceptualise complex problems in an increasingly independent, confident and flexible way and apply knowledge and skills appropriate to their solution, at the threshold of professional competence.	SC3	SCQF 10
LLO80	Justify the use of a broad selection of specialist computing and information systems skills.	SC4	SCQF 10
LLO81	Argue for the use of specialist computing and ICT facilities concerned with software development and information systems.	SC4	SCQF 10
LLO82	Communicate complex ideas to colleagues through appropriate media, based on considerations of context.	SC4	SCQF 10
LLO83	Communicate confidently in oral and formal presentations to informed audiences in a wide range of contexts.	SC4	SCQF 10
LLO84	Critique skills, knowledge and achievements in autonomous investigative, modelling and problem-solving activities.	SC5	SCQF 10
LLO85	Justify the choice of standards and quality control as applied to the specification, production, implementation and maintenance of secure information systems, in accordance with current professional and ethical codes of practices.	SC5	SCQF 10
LLO86	Critique their own knowledge and skills, enabling effective independent learning with minimal guidance, using a wide range of resources.	SC5	SCQF 10
LLO87	Justify their interaction within groups in all topics, and, where appropriate, with customers and other stakeholders.	SC5	SCQF 10

Programme Outcome Types Reference		
#	SCQF Characteristic	SCQF Level
SC1	Knowledge and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC2	Practice: Applied knowledge, skills and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC3	Generic cognitive skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10

		SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC4	Communication, numeracy and IT skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC5	Autonomy, accountability and working with others	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12

Delivery Range

<i>Title</i>	<i>Description</i>
Full Time On Campus	Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1337 Programming Bootcamp is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Stage 1 / Semester 1**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1337 Programming Bootcamp is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM1122	AI, Data and Society		Mark Bartlett (mb9400)	Yes	2.0	15	Level 7
Full Time On Campus	CM1119	Problem Solving and Maths		Mark Snaith (ms10541)	Yes	2.0	15	Level 7
Full Time On Campus	CE1337	Programming Bootcamp		Mark Snaith (ms10541)	Yes	1.0	15	Level 7
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0735 - Electives 1	Artificial Intelligence and Data Science - Electives Group 1		Laura Tucker		1.0	15	

Stage 1 / Semester 2**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1337 Programming Bootcamp is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>

Full Time On Campus	CM1123	Data Analysis		Mark Bartlett (mb9400)	Yes	3.0	15	Level 7
Full Time On Campus	CM1120	Database Systems		Tiffany Young (ty7997)	Yes	3.0	15	Level 7
Full Time On Campus	CM1121	Web UX		Jay Munro	Yes	3.0	15	Level 7
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0735 - Electives 2	Artificial Intelligence and Data Science - Electives Group 2		Laura Tucker		1.0	15	

Stage 2 / Semester 1**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1337 Programming Bootcamp is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM2137	Data Mining		Mark Bartlett (mb9400)	Yes	1.0	15	Level 8
Full Time On Campus	CE2000	Dynamic Web Technologies		David Corsar (dc9396)	Yes	1.0	15	Level 8
Full Time On Campus	CM2112	Real World Projects and Professional Skills		Ross McLean (rjm7342)	Yes	1.0	15	Level 8
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0735 - Electives 3	Artificial Intelligence and Data Science - Electives Group 3		Laura Tucker		1.0	15	

Stage 2 / Semester 2**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1337 Programming Bootcamp is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM2138	Business Intelligence		Mark Bartlett (mb9400)	Yes	1.0	15	Level 8
Full								

Time On Campus	CE2005	Data Science Development and Deployment		Mark Bartlett (mb9400)	Yes	1.0	15	Level 8
Full Time On Campus	CM2124	Research Methods		Daniel Gardner	Yes	2.0	15	Level 8
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0735 - Electives 4	Artificial Intelligence and Data Science - Electives Group 4		Laura Tucker		1.0	15	

Stage 2 / Semester 3**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1337 Programming Bootcamp is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0735 - Electives 5	Artificial Intelligence and Data Science - Electives Group 5		Laura Tucker		1.0	120	

Stage 3 / Semester 1**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1337 Programming Bootcamp is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM4125	Data Visualisation		Mark Zarb (mz7371)	Yes	1.0	15	Level 10
Full Time On Campus	CM3130	Interdisciplinary Team Project		Mark Zarb (mz7371)	Yes	1.0	15	Level 9
Full Time On Campus	CM3151	Machine Learning		David Corsar (dc9396)	Yes	1.0	15	Level 9

Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0735 - Electives 6	Artificial Intelligence and Data Science - Electives Group 6		Laura Tucker		1.0	15	

Stage 3 / Semester 2**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1337 Programming Bootcamp is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM3152	Artificial Intelligence for Problem Solving		Kit-ying Hui (cmskhu)	Yes	2.0	15	Level 9
Full Time On Campus	CM3153	Big Data Systems		Mark Bartlett (mb9400)	Yes	1.0	15	Level 9
Full Time On Campus	CM3141	Pirate Studies		Ross McLean (rjm7342)	Yes	1.0	15	Level 9
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0735 - Electives 7	Artificial Intelligence and Data Science - Electives Group 7		Laura Tucker		1.0	15	

Stage 3 / Semester 3**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1337 Programming Bootcamp is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0735 - Electives 5	Artificial Intelligence and Data Science - Electives Group 5		Laura Tucker		1.0	120	

Stage 4 / Semester 1**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1337 Programming Bootcamp is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CM4126	Computer Vision		Eyad Elyan (ee5413)	Yes	1.0	15	Level 10
Full Time On Campus	CE4145	Natural Language Processing		Mark Bartlett (mb9400)	Yes	1.0	15	Level 10
Full Time On Campus	CM4140	Optimisation		Mark Bartlett (mb9400)	Yes	1.0	15	Level 10
Elective								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	0735 - Electives 8	Artificial Intelligence and Data Science - Electives Group 8		Laura Tucker		1.0	15	

Stage 4 / Semester 2**Full Time On Campus**

Placement and Study Abroad modules provide an additional option for students, which is not required to progress to the next stage and the credits accumulated do not contribute to the award total. CM1337 Programming Bootcamp is delivered as an intensive module at the start of Semester 1 followed by the remaining Stage 1 modules undertaken in parallel over the rest of Semester 1 and Semester 2.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time On Campus	CE4321	Capstone Project		Mark Zarb (mz7371)	Yes	1.0	60	Level 10

Course Deliveries

Stage	Semester	Delivery Type	Delivery Range	Code	Title	Notes	Owner	Allow Mapping	Version	Credits	Module Level
1	Semester 1	Core	Full Time On Campus	CM1122	AI, Data and Society		Mark Bartlett (mb9400)	Yes	2.0	15	Level 7
1	Semester 1	Core	Full Time On Campus	CM1119	Problem Solving and Maths		Mark Snaith (ms10541)	Yes	2.0	15	Level 7
1	Semester 1	Core	Full Time On Campus	CE1337	Programming Bootcamp		Mark Snaith (ms10541)	Yes	1.0	15	Level 7
1	Semester 1	Elective	Full Time On Campus	0735 - Electives 1	Artificial Intelligence and Data Science - Electives Group 1		Laura Tucker		1.0	15	
1	Semester 2	Core	Full Time On Campus	CM1123	Data Analysis		Mark Bartlett (mb9400)	Yes	3.0	15	Level 7
1	Semester 2	Core	Full Time On Campus	CM1120	Database Systems		Tiffany Young (ty7997)	Yes	3.0	15	Level 7
1	Semester 2	Core	Full Time On Campus	CM1121	Web UX		Jay Munro	Yes	3.0	15	Level 7
1	Semester 2	Elective	Full Time On Campus	0735 - Electives 2	Artificial Intelligence and Data Science - Electives Group 2		Laura Tucker		1.0	15	
2	Semester 1	Core	Full Time On	CM2137	Data Mining		Mark Bartlett (mb9400)	Yes	1.0	15	Level 8

			Campus								
2	Semester 1	Core	Full Time On Campus	CE2000	Dynamic Web Technologies		David Corsar (dc9396)	Yes	1.0	15	Level 8
2	Semester 1	Core	Full Time On Campus	CM2112	Real World Projects and Professional Skills		Ross McLean (rjm7342)	Yes	1.0	15	Level 8
2	Semester 1	Elective	Full Time On Campus	0735 - Electives 3	Artificial Intelligence and Data Science - Electives Group 3		Laura Tucker		1.0	15	
2	Semester 2	Core	Full Time On Campus	CM2138	Business Intelligence		Mark Bartlett (mb9400)	Yes	1.0	15	Level 8
2	Semester 2	Core	Full Time On Campus	CE2005	Data Science Development and Deployment		Mark Bartlett (mb9400)	Yes	1.0	15	Level 8
2	Semester 2	Core	Full Time On Campus	CM2124	Research Methods		Daniel Gardner	Yes	2.0	15	Level 8
2	Semester 2	Elective	Full Time On Campus	0735 - Electives 4	Artificial Intelligence and Data Science - Electives Group 4		Laura Tucker		1.0	15	
2	Semester 3	Elective	Full Time On Campus	0735 - Electives 5	Artificial Intelligence and Data Science - Electives Group 5		Laura Tucker		1.0	120	
3	Semester 1	Core	Full Time On Campus	CM4125	Data Visualisation		Mark Zarb (mz7371)	Yes	1.0	15	Level 10
			Full								

3	Semester 1	Core	Time On Campus	CM3130	Interdisciplinary Team Project		Mark Zarb (mz7371)	Yes	1.0	15	Level 9
3	Semester 1	Core	Full Time On Campus	CM3151	Machine Learning		David Corsar (dc9396)	Yes	1.0	15	Level 9
3	Semester 1	Elective	Full Time On Campus	0735 - Electives 6	Artificial Intelligence and Data Science - Electives Group 6		Laura Tucker		1.0	15	
3	Semester 2	Core	Full Time On Campus	CM3152	Artificial Intelligence for Problem Solving		Kit-ying Hui (cmskhu)	Yes	2.0	15	Level 9
3	Semester 2	Core	Full Time On Campus	CM3153	Big Data Systems		Mark Bartlett (mb9400)	Yes	1.0	15	Level 9
3	Semester 2	Core	Full Time On Campus	CM3141	Pirate Studies		Ross McLean (rjm7342)	Yes	1.0	15	Level 9
3	Semester 2	Elective	Full Time On Campus	0735 - Electives 7	Artificial Intelligence and Data Science - Electives Group 7		Laura Tucker		1.0	15	
3	Semester 3	Elective	Full Time On Campus	0735 - Electives 5	Artificial Intelligence and Data Science - Electives Group 5		Laura Tucker		1.0	120	
4	Semester 1	Core	Full Time On Campus	CM4126	Computer Vision		Eyad Elyan (ee5413)	Yes	1.0	15	Level 10
4	Semester 1	Core	Full Time On Campus	CE4145	Natural Language Processing		Mark Bartlett (mb9400)	Yes	1.0	15	Level 10

			Campus								
4	Semester 1	Core	Full Time On Campus	CM4140	Optimisation		Mark Bartlett (mb9400)	Yes	1.0	15	Level 10
4	Semester 1	Elective	Full Time On Campus	0735 - Electives 8	Artificial Intelligence and Data Science - Electives Group 8		Laura Tucker		1.0	15	
4	Semester 2	Core	Full Time On Campus	CE4321	Capstone Project		Mark Zarb (mz7371)	Yes	1.0	60	Level 10

Delivery

Location	Mode of Study	Mode of Attendance	Teaching Period Key Date
Aberdeen, United Kingdom	On-Campus	Full Time	

Sources of Further Information

#	URL and Description
1	https://www.bcs.org/media/1209/accreditation-guidelines.pdf The accreditation guidelines for the BCS, The Chartered Institute for IT, Academic Accreditation Guidelines, April 2022.
2	https://www.acm.org/binaries/content/assets/education/curricula-recommendations/dstf_ccdsc2021.pdf The Association for Computing Machinery (ACM) Data Science Task Force's Computing Competencies for Undergraduate Data Science Curricula, January 2021